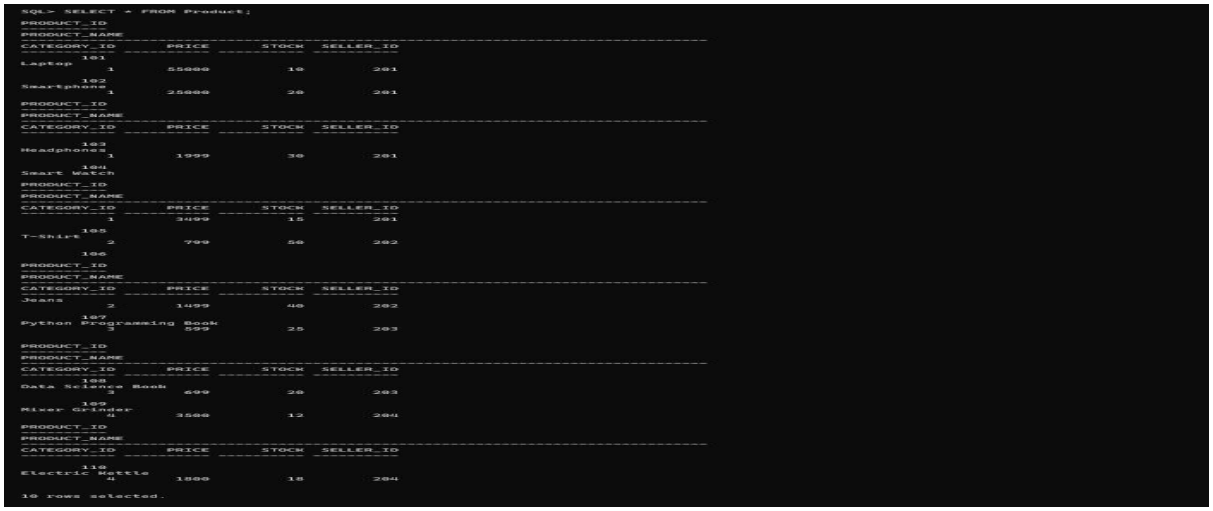
Amazon E-Commerce Database Management System

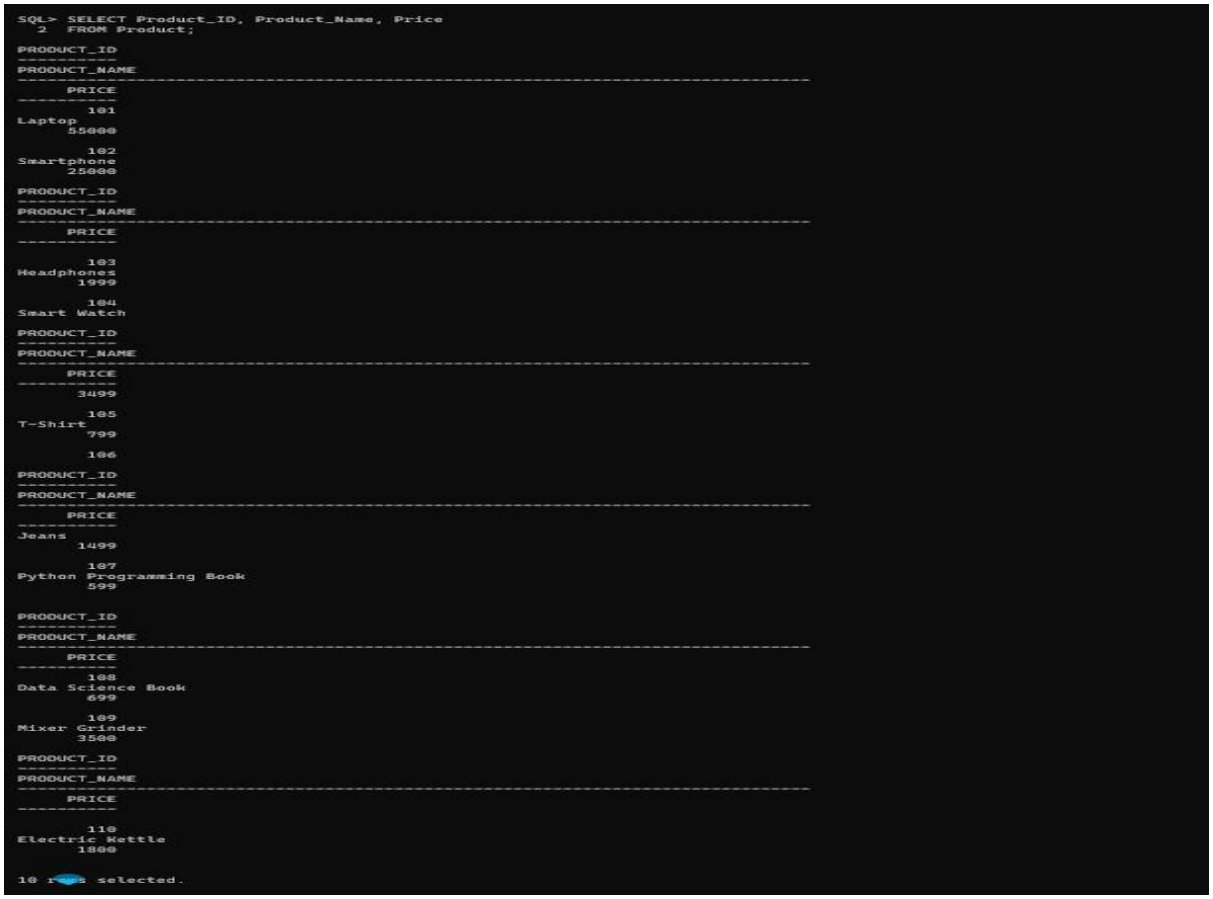
CHAPTER – 8

1. Basic SQL Queries

1.1 SELECT Query — Display all products



1.2 Display selected product details



1.3 WHERE Clause — Find products in Category 1

```
SQL> SELECT Product_ID, Product_Name
  2  FROM Product
  3  WHERE Category_ID = 1;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
       101
Laptop

       102
Smartphone

       103
Headphones


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
       104
Smart Watch
```

1.4 ORDER BY — Display products by descending price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Laptop
     55000

       102
Smartphone
     25000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       109
Mixer Grinder
      3500

       104
Smart Watch

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      3499

       103
Headphones
      1999

       110

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Electric Kettle
      1800

       106
Jeans
      1499


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       105
T-Shirt
       799

       108
Data Science Book
       699

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       107
Python Programming Book
       599

10 rows selected.
```

1.5 DISTINCT — Display unique category IDs

```
SQL> SELECT DISTINCT Category_ID
  2  FROM Product;S
  3
SQL> SELECT DISTINCT Category_ID
  2  FROM Product;

CATEGORY_ID
-----------
          1
          2
          3
          4
```

2. Search Products

2.1 Search products based on price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price < 2000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       103
Headphones
      1999

       105
T-Shirt
       799

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       106
Jeans
      1499

       107
Python Programming Book

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       599

       108
Data Science Book
       699

       110

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Electric Kettle
      1800


6 rows selected.
```

2.2 Search products based on category

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Category_ID = 2;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       105
T-Shirt
       799

       106
Jeans
      1499

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
```

2.3 Search available products

```
SQL> SELECT
  2      p.Product_ID,
  3      p.Product_Name,
  4      i.Available_Stock
  5  FROM Product p
  6  JOIN Inventory i
  7  ON p.Product_ID = i.Product_ID
  8  WHERE i.Available_Stock > 0;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
AVAILABLE_STOCK
---------------
       101
Laptop
              10

       102
Smartphone
              20

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
AVAILABLE_STOCK
---------------

       103
Headphones
              30

       104
Smart Watch
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
AVAILABLE_STOCK
---------------
              15

       105
T-Shirt
              50

       106

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
AVAILABLE_STOCK
---------------
Jeans
              40

       107
Python Programming Book
              25

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
AVAILABLE_STOCK
---------------
       109
Mixer Grinder
              12

8 rows selected.
```

3. Retrieve Customer Information

3.1 Display all customers



3.2 Display selected customer details

```
SQL> SELECT Customer_ID, First_Name, Last_Name
  2  FROM Customer;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
        101 Arun
Kumar

        102 Priya
Sharma

        103 Rahul
Verma


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
        104 Sneha
Reddy

        105 Karthik
Raj

        106 Divya
Nair


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
        107 Vignesh
Mohan


7 rows selected.
```

4. Apply Filtering Conditions

4.1 AND condition

```
SQL> SELECT Product_ID, Product_Name, Price, Stock
  2  FROM Product
  3  WHERE Price > 1000 AND Stock < 20;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
       101
Laptop
     55000         10

       104
Smart Watch
      3499         15

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

       109
Mixer Grinder
      3500         12

       110
Electric Kettle

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
      1800         18
```

4.2 OR condition

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price < 1000 OR Price > 5000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Laptop
     55000

       102
Smartphone
     25000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       105
T-Shirt
       799

       107
Python Programming Book

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       599

       108
Data Science Book
       699
```

4.3 NOT condition

```
SQL> 4.3 NOT condition
SP2-0734: unknown command beginning "4.3 NOT co..." - rest of line ignored.
SQL> SELECT Product_ID, Product_Name, Category_ID
  2  FROM Product
  3  WHERE NOT Category_ID = 1;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID
-----------
       105
T-Shirt
           2

       106
Jeans
           2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID
-----------

       107
Python Programming Book
           3

       108
Data Science Book

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID
-----------
           3

       109
Mixer Grinder
           4

       110

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID
-----------
Electric Kettle
           4


6 rows selected.
```

5. Basic Business Reports

5.1 Product Price Report

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Laptop
     55000

       102
Smartphone
     25000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       109
Mixer Grinder
      3500

       104
Smart Watch
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      3499

       103
Headphones
      1999

       110

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Electric Kettle
      1800

       106
Jeans
      1499

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       105
T-Shirt
       799

       108
Data Science Book
       499

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       107
Python Programming Book
       599

10 rows selected.
```

5.2 Customer Order Report

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name,
  4      c.Last_Name,
  5      o.Order_ID,
  6      o.Order_Date,
  7      o.Total_Amount,
  8      o.Order_Status
  9  FROM Customer c
 10  JOIN Orders o
 11  ON c.Customer_ID = o.Customer_ID
 12  ORDER BY c.Customer_ID;

no rows selected
```

5.3 Product Order Report

```
SQL> SELECT
  2      p.Product_ID,
  3      p.Product_Name,
  4      oi.Order_ID,
  5      oi.Quantity,
  6      oi.Unit_Price
  7  FROM Product p
  8  JOIN Order_Item oi
  9  ON p.Product_ID = oi.Product_ID
 10  ORDER BY oi.Order_ID;

no rows selected
```

5.4 Customer-wise Total Orders

```
SQL> SELECT
  2       c.Customer_ID,
  3       c.First_Name,
  4       c.Last_Name,
  5       COUNT(o.Order_ID) AS Total_Orders
  6  FROM Customer c
  7  LEFT JOIN Orders o
  8  ON c.Customer_ID = o.Customer_ID
  9  GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
 10  ORDER BY Total_Orders DESC;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS
-------------------------------------------------- ------------
        107 Vignesh
Mohan                                                         0

        105 Karthik
Raj                                                           0

        104 Sneha
Reddy                                                         0

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS
-------------------------------------------------- ------------
        106 Divya
Nair                                                          0

        103 Rahul
Verma                                                         0

        102 Priya
Sharma                                                        0

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS
-------------------------------------------------- ------------
        101 Arun
Kumar                                                         0

7 rows selected.
```

5.5 Customer-wise Total Purchase Amount

```
SQL> SELECT
  2       c.Customer_ID,
  3       c.First_Name,
  4       c.Last_Name,
  5       NVL(SUM(o.Total_Amount), 0) AS Total_Purchase_Amount
  6  FROM Customer c
  7  LEFT JOIN Orders o
  8  ON c.Customer_ID = o.Customer_ID
  9  GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
 10  ORDER BY Total_Purchase_Amount DESC;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_PURCHASE_AMOUNT
-------------------------------------------------- ---------------------
        107 Vignesh
Mohan                                                                 0

        105 Karthik
Raj                                                                   0

        104 Sneha
Reddy                                                                 0


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_PURCHASE_AMOUNT
-------------------------------------------------- ---------------------
        106 Divya
Nair                                                                  0

        103 Rahul
Verma                                                                 0

        102 Priya
Sharma                                                                0


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_PURCHASE_AMOUNT
-------------------------------------------------- ---------------------
        101 Arun
Kumar                                                                 0


7 rows selected.
```